EN.553.632 Bayesian Statistics

Jonathan Y. Ma

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Chapter 1

Introduction and Probability Review

1.1 Bayesian vs Frequentist Perspectives

Definition 1.1.1. Probability can be interpreted in two fundamentally different ways:

- **Frequentist:** Probability is the long-run relative frequency of an event.
- **Bayesian:** Probability is a degree of belief about an uncertain quantity.

In the Bayesian framework, unknown parameters are treated as random variables. This contrasts with the frequentist view, where parameters are fixed but unknown.

Example 1.1.1. Consider a coin flipped $n = 10$ times with 7 heads observed.

- Frequentist estimate: $\hat{\theta} = 0.7$
- Bayesian approach: place a prior $\theta \sim \text{Beta}(\alpha, \beta)$

After observing $y = 7$ heads, the posterior becomes:

$$\theta \mid y \sim \text{Beta}(\alpha + y, \beta + n - y)$$

1.2 Bayes' Theorem

Theorem 1.2.1 (Bayes' Rule)**.** *For parameter θ and data y,*

$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)}$$

where

$$p(y) = \int p(y \mid \theta)p(\theta)\,\mathrm{d}\theta$$

Remark 1.2.1. The posterior is often written up to proportionality:

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$$

1.3 Bernoulli and Binomial Models

1.3.1 Bernoulli Model

Definition 1.3.1. A Bernoulli random variable satisfies:

$$Y \sim \text{Bernoulli}(\theta), \quad P(Y = 1) = \theta$$

1.3.2 Binomial Model

Definition 1.3.2. For n independent Bernoulli trials,

$$Y \sim \text{Binomial}(n, \theta), \quad P(Y = y|\theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}$$

1.3.3 Likelihood

Given observations $y_1, \ldots, y_n$, the likelihood is:

$$L(\theta) = \prod_{i=1}^{n} \theta^{y_i}(1-\theta)^{1-y_i} = \theta^{\sum y_i}(1-\theta)^{n-\sum y_i}$$

1.3.4 Maximum Likelihood Estimation

Proposition 1.3.1. *The MLE for θ in the Bernoulli/Binomial model is:*

$$\hat{\theta}_{MLE} = \bar{y}$$

Proof. The log-likelihood is:

$$\ell(\theta) = \sum y_i \log\theta + (n - \sum y_i)\log(1-\theta)$$

Differentiate:

$$\frac{d\ell}{d\theta} = \frac{\sum y_i}{\theta} - \frac{n - \sum y_i}{1-\theta}$$

Set equal to zero:

$$\frac{\sum y_i}{\theta} = \frac{n - \sum y_i}{1-\theta} \Rightarrow \theta = \frac{1}{n}\sum y_i$$

□

1.4 Conjugate Priors: Beta-Binomial Model

Definition 1.4.1. A prior is conjugate if the posterior belongs to the same family as the prior.

Proposition 1.4.1. *If*

$$\theta \sim \text{Beta}(\alpha, \beta), \quad Y \mid \theta \sim \text{Binomial}(n, \theta)$$

then

$$\theta \mid y \sim \text{Beta}(\alpha + y, \beta + n - y)$$

Proof. Posterior:

$$p(\theta \mid y) \propto \theta^y (1-\theta)^{n-y} \cdot \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

Combine exponents:

$$p(\theta \mid y) \propto \theta^{\alpha+y-1}(1-\theta)^{\beta+n-y-1}$$

which is the kernel of a Beta distribution. □

1.4.1 Posterior Mean

$$\mathbb{E}[\theta \mid y] = \frac{\alpha + y}{\alpha + \beta + n}$$

Remark 1.4.1. This can be interpreted as a weighted average:

$$\mathbb{E}[\theta \mid y] = w \cdot \frac{\alpha}{\alpha + \beta} + (1 - w) \cdot \frac{y}{n}$$

1.5 Credible Intervals and HPD Regions

Definition 1.5.1. An interval (a, b) is a $(1 - \alpha)$ credible interval if

$$\mathbb{P}(a < \theta < b \mid y) = 1 - \alpha$$

Definition 1.5.2. A highest posterior density (HPD) region A satisfies:

- $\mathbb{P}(\theta \in A \mid y) = 1 - \alpha$
- $p(\theta_1 \mid y) > p(\theta_2 \mid y)$ for $\theta_1 \in A$, $\theta_2 \notin A$

1.6 Posterior Predictive Distribution

Proposition 1.6.1. *For a new observation Z,*

$$p(Z = 1 \mid y) = \mathbb{E}[\theta \mid y]$$

Proof.

$$p(Z = 1 \mid y) = \int p(Z = 1 \mid \theta) p(\theta \mid y) \, \mathrm{d}\theta = \int \theta p(\theta \mid y) \, \mathrm{d}\theta = \mathbb{E}[\theta \mid y]$$

□

1.7 Exponential Family

Definition 1.7.1. A distribution belongs to the exponential family if it can be written as:

$$p(y \mid \theta) = f(y) g(\theta) \exp\{\phi(\theta) u(y)\}$$

Remark 1.7.1. Many common distributions (Normal, Binomial, Poisson) belong to this family, which enables conjugate prior constructions.

1.8 Prior Specification

1.8.1 Subjective Priors

Constructed from expert knowledge.

1.8.2 Non-informative Priors

Designed to have minimal influence on the posterior.

Example 1.8.1. A flat prior:

$$\pi(\theta) \propto 1$$

1.8.3 Jeffreys Prior

Definition 1.8.1.

$$\pi(\theta) \propto \sqrt{I(\theta)}, \quad I(\theta) = -\mathbb{E}\left[\frac{\partial^2}{\partial \theta^2} \log p(y \mid \theta)\right]$$

Remark 1.8.1. Jeffreys priors are invariant under reparameterization.

1.9 Poisson-Gamma Model

1.9.1 Model

$$Y_i \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$$

Likelihood:

$$L(\theta) \propto \theta^{\sum y_i} e^{-n\theta}$$

1.9.2 Prior

$$\theta \sim \text{Gamma}(a, b)$$

1.9.3 Posterior

Proposition 1.9.1.

$$\theta \mid y \sim \text{Gamma}(a + \sum y_i, b + n)$$

1.10 Preview: Computational Methods

In many models, posterior distributions cannot be computed analytically. This motivates:

- Monte Carlo integration
- Importance sampling
- Markov Chain Monte Carlo (MCMC)

These methods will be developed in later chapters.

1.11 Common Distributions and Identities

1.11.1 Beta and Gamma Functions

Definition 1.11.1. The Gamma function is defined as:

$$\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} \, \mathrm{d}x$$

and satisfies:

$$\Gamma(n) = (n-1)! \quad \text{for integer } n$$

Definition 1.11.2. The Beta distribution has density:

$$f(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1}(1-\theta)^{b-1}$$

Proposition 1.11.1. *For $\theta \sim \text{Beta}(a, b)$:*

$$\mathbb{E}[\theta] = \frac{a}{a+b}, \quad \text{Var}(\theta) = \frac{ab}{(a+b)^2(a+b+1)}$$

1.11.2 Gamma Distribution

Definition 1.11.3. A Gamma distribution with shape a and rate b:

$$f(\theta) = \frac{b^a}{\Gamma(a)}\theta^{a-1}e^{-b\theta}$$

Proposition 1.11.2.

$$\mathbb{E}[\theta] = \frac{a}{b}, \quad \text{Var}(\theta) = \frac{a}{b^2}$$

Remark 1.11.1. The Gamma distribution is conjugate to the Poisson likelihood.

1.12 Sufficiency and Likelihood Structure

Definition 1.12.1. A statistic $T(y)$ is sufficient for θ if the likelihood can be written as:

$$p(y \mid \theta) = g(T(y), \theta)h(y)$$

Proposition 1.12.1. *For:*

- *Binomial model: sufficient statistic is $y = \sum y_i$*
- *Poisson model: sufficient statistic is $\sum y_i$*

Remark 1.12.1. This reduction is critical in Bayesian updating since posterior distributions depend only on sufficient statistics.

1.13 Posterior as a Weighted Average

A key Bayesian insight is that posterior estimates combine prior and data:

Proposition 1.13.1. *For the Beta-Binomial model:*

$$\mathbb{E}[\theta \mid y] = \frac{a+y}{a+b+n}$$

can be written as:

$$\mathbb{E}[\theta \mid y] = \frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \frac{y}{n}$$

Remark 1.13.1. This shows:

- Prior contributes as pseudo-observations
- Data influence increases with n

1.14 Bias, Variance, and MSE

Definition 1.14.1. Bias of an estimator $\hat{\theta}$:

$$\text{Bias} = \mathbb{E}[\hat{\theta}] - \theta$$

Definition 1.14.2. Mean Squared Error:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}^2$$

Remark 1.14.1. Bayesian estimators are often biased but achieve lower MSE through shrinkage.

1.15 Poisson-Gamma Model (Detailed)

1.15.1 Likelihood

$$L(\theta) = \prod_{i=1}^{n} \frac{\theta^{y_i} e^{-\theta}}{y_i!} \propto \theta^{\sum y_i} e^{-n\theta}$$

1.15.2 Posterior Derivation

Proof. With prior $\theta \sim \text{Gamma}(a, b)$:

$$p(\theta \mid y) \propto \theta^{\sum y_i} e^{-n\theta} \cdot \theta^{a-1} e^{-b\theta}$$

$$= \theta^{a+\sum y_i - 1} e^{-(b+n)\theta}$$

which is $\text{Gamma}(a + \sum y_i, b + n)$. □

1.15.3 Posterior Mean Interpretation

$$\mathbb{E}[\theta \mid y] = \frac{b}{b+n} \cdot \frac{a}{b} + \frac{n}{b+n} \cdot \frac{\sum y_i}{n}$$

Remark 1.15.1. Again, this is a weighted average of prior mean and sample mean.

1.16 Posterior Predictive Distribution

Definition 1.16.1. The posterior predictive distribution is:

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta) p(\theta \mid y) \, \mathrm{d}\theta$$

Remark 1.16.1. This integrates out parameter uncertainty, a key advantage over plug-in estimators.

1.17 Exponential Family and Conjugacy

Definition 1.17.1. A one-parameter exponential family has the form:

$$p(y \mid \phi) = h(y)c(\phi)\exp\{\phi t(y)\}$$

Proposition 1.17.1. *For exponential families:*

- *Sufficient statistics appear naturally as $t(y)$*
- *Conjugate priors exist due to exponential structure*

Remark 1.17.1. Examples:

- Binomial $\leftrightarrow$ Beta
- Poisson $\leftrightarrow$ Gamma
- Normal $\leftrightarrow$ Normal / Inverse-Gamma

1.18 Interpretation of Bayesian Updating

Remark 1.18.1. Bayesian inference provides:

- Full uncertainty quantification (posterior distribution)
- Direct probabilistic statements about parameters
- Natural predictive distributions

Example 1.18.1. In rare-event estimation, even if $y = 0$, the posterior remains well-defined:

$$\theta \mid y \sim \text{Beta}(a, b + n)$$

which avoids degenerate estimates like $\hat{\theta} = 0$.

Chapter 2

The Binomial Model

2.1 Model Setup

Definition 2.1.1. Let $Y \mid \theta \sim \text{Binomial}(n, \theta)$ with probability mass function:

$$p(y \mid \theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}, \quad \theta \in [0, 1]$$

Remark 2.1.1. If $Y_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$, then $Y = \sum_{i=1}^{n} Y_i \sim \text{Binomial}(n, \theta)$. The sufficient statistic is $y = \sum Y_i$.

2.2 Likelihood and Maximum Likelihood Estimation

Ignoring constants:

$$L(\theta) \propto \theta^y (1-\theta)^{n-y}$$

Proposition 2.2.1. *The log-likelihood is:*

$$\ell(\theta) = y \log \theta + (n-y) \log(1-\theta)$$

Proposition 2.2.2. *The MLE for θ is:*

$$\hat{\theta} = \frac{y}{n}$$

Proof. Differentiate:

$$\frac{d\ell}{d\theta} = \frac{y}{\theta} - \frac{n-y}{1-\theta}$$

Set equal to zero:

$$\frac{y}{\theta} = \frac{n-y}{1-\theta} \Rightarrow \theta = \frac{y}{n}$$

□

2.3 Beta Prior

Definition 2.3.1. Assume prior:

$$\theta \sim \text{Beta}(a, b), \quad \pi(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}$$

Remark 2.3.1. The parameters (a, b) can be interpreted as prior pseudo-counts:

$$a-1 \text{ successes}, \quad b-1 \text{ failures}$$

2.4 Posterior Distribution

Theorem 2.4.1. *If* $\theta \sim \text{Beta}(a, b)$ *and* $Y \mid \theta \sim \text{Binomial}(n, \theta)$*, then:*

$$\theta \mid y \sim \text{Beta}(a + y, b + n - y)$$

Proof.

$$p(\theta \mid y) \propto \theta^y (1-\theta)^{n-y} \cdot \theta^{a-1}(1-\theta)^{b-1}$$

$$= \theta^{a+y-1}(1-\theta)^{b+n-y-1}$$

□

2.5 Posterior Summaries

2.5.1 Mean

$$\mathbb{E}[\theta \mid y] = \frac{a+y}{a+b+n}$$

Remark 2.5.1. This can be written as:

$$\mathbb{E}[\theta \mid y] = \frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \frac{y}{n}$$

2.5.2 Mode

$$\text{Mode}(\theta \mid y) = \frac{a+y-1}{a+b+n-2}$$

2.5.3 Variance

$$\text{Var}(\theta \mid y) = \frac{(a+y)(b+n-y)}{(a+b+n)^2(a+b+n+1)}$$

2.6 Posterior Predictive Distribution

Definition 2.6.1. For a new observation $\tilde{Y}$:

$$p(\tilde{Y} = 1 \mid y) = \int \theta p(\theta \mid y) \, \mathrm{d}\theta$$

Proposition 2.6.1.

$$p(\tilde{Y} = 1 \mid y) = \mathbb{E}[\theta \mid y] = \frac{a + y}{a + b + n}$$

2.7 Credible Intervals

Definition 2.7.1. An interval (l, u) is a $(1 - \alpha)$ credible interval if:

$$\mathbb{P}(l < \theta < u \mid y) = 1 - \alpha$$

2.8 Highest Posterior Density Regions

Definition 2.8.1. A set A is a HPD region if:

- $\mathbb{P}(\theta \in A \mid y) = 1 - \alpha$
- $p(\theta_1 \mid y) > p(\theta_2 \mid y)$ for $\theta_1 \in A$, $\theta_2 \notin A$

2.9 Shrinkage Interpretation

Proposition 2.9.1. *The posterior mean can be written as:*

$$\mathbb{E}[\theta \mid y] = w \cdot \frac{a}{a + b} + (1 - w) \cdot \frac{y}{n}$$

where:

$$w = \frac{a + b}{a + b + n}$$

Remark 2.9.1.

- Small n: prior dominates
- Large n: data dominates

2.10 Rare Event Example

Example 2.10.1. Suppose:

$$n = 20, \quad y = 0, \quad \theta \sim \text{Beta}(2, 20)$$

Then:

$$\theta \mid y \sim \text{Beta}(2, 40)$$

Remark 2.10.1. The MLE gives $\hat{\theta} = 0$, while the Bayesian estimator remains strictly positive.

2.11 Frequentist vs Bayesian Interpretation

Remark 2.11.1. Frequentist interval:

$$\mathbb{P}(l(Y) < \theta < u(Y) \mid \theta) = 1 - \alpha$$

Bayesian interval:

$$\mathbb{P}(l < \theta < u \mid y) = 1 - \alpha$$

Chapter 3

Poisson Model and Exponential Families

3.1 Poisson Model

Definition 3.1.1. A random variable Y follows a Poisson distribution with rate $\theta > 0$ if:

$$Y \mid \theta \sim \text{Poisson}(\theta), \quad p(y \mid \theta) = \frac{\theta^y e^{-\theta}}{y!}, \quad y \in \mathbb{N}$$

Proposition 3.1.1. *For $Y \sim \text{Poisson}(\theta)$:*

$$\mathbb{E}[Y] = \theta, \quad \text{Var}(Y) = \theta$$

Remark 3.1.1. The equality of mean and variance is a defining feature of the Poisson model. Large deviations between sample mean and variance indicate model misspecification.

3.2 Likelihood and Sufficiency

Let $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$. The joint likelihood is:

$$p(y_1, \ldots, y_n \mid \theta) = \prod_{i=1}^{n} \frac{\theta^{y_i} e^{-\theta}}{y_i!} = c(y)\, \theta^{\sum y_i} e^{-n\theta}$$

Proposition 3.2.1. *The sufficient statistic is:*

$$T(y) = \sum_{i=1}^{n} y_i$$

Remark 3.2.1. All information about θ is contained in the total count $\sum y_i$.

3.3 Maximum Likelihood Estimation

Proposition 3.3.1. *The MLE for θ is:*

$$\hat{\theta} = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

Proof. Log-likelihood:

$$\ell(\theta) = \sum y_i \log\theta - n\theta$$

Differentiate:

$$\frac{d\ell}{d\theta} = \frac{\sum y_i}{\theta} - n$$

Set to zero:

$$\theta = \frac{1}{n}\sum y_i$$

□

3.4 Gamma Prior

Definition 3.4.1. Assume prior:

$$\theta \sim \text{Gamma}(a,b), \quad \pi(\theta) = \frac{b^a}{\Gamma(a)}\theta^{a-1}e^{-b\theta}$$

Proposition 3.4.1.

$$\mathbb{E}[\theta] = \frac{a}{b}, \quad \text{Var}(\theta) = \frac{a}{b^2}$$

3.5 Posterior Distribution

Theorem 3.5.1. *If $\theta \sim \text{Gamma}(a,b)$ and $Y_i \overset{iid}{\sim} \text{Poisson}(\theta)$, then:*

$$\theta \mid y \sim \text{Gamma}\left(a + \sum y_i,\ b + n\right)$$

Proof.

$$p(\theta \mid y) \propto \theta^{\sum y_i} e^{-n\theta} \cdot \theta^{a-1} e^{-b\theta}$$

$$= \theta^{a+\sum y_i - 1} e^{-(b+n)\theta}$$

□

3.6 Posterior Summaries

3.6.1 Mean

$$\mathbb{E}[\theta \mid y] = \frac{a + \sum y_i}{b + n}$$

Remark 3.6.1. This can be written as:

$$\mathbb{E}[\theta \mid y] = \frac{b}{b+n} \cdot \frac{a}{b} + \frac{n}{b+n} \cdot \frac{\sum y_i}{n}$$

3.6.2 Variance

$$\mathrm{Var}(\theta \mid y) = \frac{a + \sum y_i}{(b+n)^2}$$

3.7 Posterior Predictive Distribution

Definition 3.7.1. For a new observation $\tilde{Y}$:

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta) p(\theta \mid y) \, \mathrm{d}\theta$$

Remark 3.7.1. This results in a negative binomial distribution, reflecting overdispersion relative to the Poisson model.

3.8 Example: Birth Rate Comparison

Example 3.8.1. Suppose:

- Group 1: $n_1 = 111$, $\sum y_i = 217$
- Group 2: $n_2 = 44$, $\sum y_i = 66$
- Prior: $\theta \sim \mathrm{Gamma}(2, 1)$

Posteriors:

$$\theta_1 \mid y \sim \mathrm{Gamma}(219, 112), \quad \theta_2 \mid y \sim \mathrm{Gamma}(68, 45)$$

3.9 Exponential Family Representation

Definition 3.9.1. A distribution belongs to the exponential family if:

$$p(y \mid \phi) = h(y) c(\phi) \exp\{\phi t(y)\}$$

3.9.1 Poisson as an Exponential Family

Proposition 3.9.1. *The Poisson distribution can be written as:*

$$p(y \mid \theta) = \frac{1}{y!} \exp\{y \log \theta - \theta\}$$

Remark 3.9.1. This corresponds to:

$$t(y) = y, \quad \phi = \log \theta$$

3.10 General Exponential Family Structure

Definition 3.10.1. A one-parameter exponential family has the form:

$$p(y \mid \phi) = h(y)c(\phi) \exp\{\phi t(y)\}$$

Proposition 3.10.1. *For exponential families:*

- $t(y)$ *is a sufficient statistic*
- *Conjugate priors exist due to exponential structure*

Remark 3.10.1. Common examples:

- Binomial $\leftrightarrow$ Beta
- Poisson $\leftrightarrow$ Gamma
- Normal $\leftrightarrow$ Normal / Inverse-Gamma

3.11 Interpretation and Model Limitations

Remark 3.11.1. The Poisson model assumes:

- independence of events
- constant rate θ
- equal mean and variance

Remark 3.11.2. If variance exceeds the mean, a negative binomial model may be more appropriate.

Chapter 4

Monte Carlo Methods

4.1 Motivation

Remark 4.1.1. In many Bayesian models, posterior quantities cannot be computed analytically. Instead, we approximate expectations using simulation methods.

Definition 4.1.1. We are often interested in computing:

$$\mathbb{E}[f(\theta)] = \int f(\theta)p(\theta)\,d\theta$$

Remark 4.1.2. Monte Carlo methods approximate this integral using random samples.

4.2 Monte Carlo Estimation

Definition 4.2.1. Let $\theta^{(1)}, \ldots, \theta^{(S)} \sim p(\theta)$ be i.i.d. samples. Then:

$$\hat{\mu} = \frac{1}{S}\sum_{s=1}^{S} f(\theta^{(s)})$$

Proposition 4.2.1. *By the Law of Large Numbers:*

$$\hat{\mu} \to \mathbb{E}[f(\theta)] \quad \textit{as } S \to \infty$$

4.2.1 Variance and Error

Definition 4.2.2. Monte Carlo variance:

$$\widehat{\mathrm{Var}} = \frac{1}{S-1}\sum_{s=1}^{S}\left(f(\theta^{(s)}) - \hat{\mu}\right)^2$$

Definition 4.2.3. Monte Carlo standard error:

$$\text{MCSE} = \sqrt{\frac{\widehat{\text{Var}}}{S}}$$

Definition 4.2.4. Monte Carlo interval:

$$\hat{\mu} \pm 2 \cdot \text{MCSE}$$

4.3 Importance Sampling

Remark 4.3.1. Direct sampling from $p(\theta)$ is often difficult. Instead, we sample from an easier distribution $g(\theta)$.

Definition 4.3.1. Rewrite the expectation:

$$\mathbb{E}[f(\theta)] = \int f(\theta)\frac{p(\theta)}{g(\theta)}g(\theta)\,d\theta$$

Definition 4.3.2. Define weights:

$$w(\theta) = \frac{p(\theta)}{g(\theta)}$$

4.3.1 Estimators

- Unnormalized:

$$\hat{\mu}^* = \frac{1}{S}\sum_{s=1}^{S} f(\theta^{(s)})w^*(\theta^{(s)})$$

- Normalized:

$$\hat{\mu} = \sum_{s=1}^{S} f(\theta^{(s)})w(\theta^{(s)})$$

  where:

$$w(\theta^{(s)}) = \frac{w^*(\theta^{(s)})}{\sum_{j=1}^{S} w^*(\theta^{(j)})}$$

4.3.2 Effective Sample Size

Definition 4.3.3. Effective sample size (ESS):

$$\text{ESS} \approx \frac{S}{1 + \text{Var}(w^*)}$$

Remark 4.3.2. High variance in weights reduces efficiency.

4.4 Accept-Reject Sampling

Definition 4.4.1. Suppose we want to sample from $p(x)$ using proposal $g(x)$ such that:

$$p(x) \leq Mg(x)$$

4.4.1 Algorithm

1. Sample $X \sim g(x)$
2. Sample $U \sim \text{Uniform}(0, 1)$
3. Accept if:
$$U \leq \frac{p(X)}{Mg(X)}$$
4. Otherwise reject and repeat

Remark 4.4.1. Efficiency depends on choosing M as small as possible.

4.5 Limitations of Direct Methods

Remark 4.5.1. Monte Carlo methods require:

- ability to sample from target distribution
- tractable density evaluation

Remark 4.5.2. When this is not possible, we use Markov Chain Monte Carlo (MCMC).

4.6 Markov Chains

Definition 4.6.1. A Markov chain satisfies:

$$P(X_{n+1} \mid X_n, \ldots, X_0) = P(X_{n+1} \mid X_n)$$

Definition 4.6.2. A stationary distribution π satisfies:

$$\pi = \pi P$$

Remark 4.6.1. We construct chains whose stationary distribution is the posterior $p(\theta|y)$.

4.7 Markov Chain Monte Carlo (MCMC)

Definition 4.7.1. MCMC methods generate dependent samples $\theta^{(1)}, \ldots, \theta^{(S)}$ such that:

$$\theta^{(s)} \sim p(\theta \mid y) \quad \text{as } s \to \infty$$

Proposition 4.7.1. *Ergodic theorem:*

$$\frac{1}{S}\sum_{s=1}^{S} f(\theta^{(s)}) \to \mathbb{E}[f(\theta \mid y)]$$

Remark 4.7.1. Samples are no longer independent, which affects variance.

4.8 Metropolis-Hastings Algorithm

Definition 4.8.1. Given proposal $q(\theta^* \mid \theta)$, define acceptance probability:

$$\alpha(\theta, \theta^*) = \min\left\{\frac{p(\theta^* \mid y)q(\theta \mid \theta^*)}{p(\theta \mid y)q(\theta^* \mid \theta)}, 1\right\}$$

4.8.1 Algorithm

1. Initialize $\theta^{(0)}$
2. Propose $\theta^* \sim q(\cdot \mid \theta^{(t-1)})$
3. Accept with probability α
4. Otherwise keep $\theta^{(t)} = \theta^{(t-1)}$

Remark 4.8.1. If q is symmetric, the ratio simplifies.

4.9 Random Walk Metropolis

Definition 4.9.1. Proposal:

$$\theta^* = \theta^{(t-1)} + \epsilon$$

Remark 4.9.1. Common choices for ϵ:

- Normal
- Student-t
- Uniform

4.10 Gibbs Sampling

Definition 4.10.1. Instead of sampling jointly, we sample from conditionals:

$$\theta_1 \sim p(\theta_1 \mid \theta_2, y), \quad \theta_2 \sim p(\theta_2 \mid \theta_1, y)$$

Remark 4.10.1. This is especially useful in high-dimensional problems.

4.11 Burn-in and Convergence

Definition 4.11.1. Burn-in: discard first B samples:

$$\{\theta^{(B+1)}, \ldots, \theta^{(B+S)}\}$$

Remark 4.11.1. Early samples depend on initialization and may not represent the target distribution.

4.12 Autocorrelation and Efficiency

Definition 4.12.1. Autocorrelation at lag t:

$$\rho_t = \frac{\sum_{s=1}^{S-t}(\theta_s - \bar{\theta})(\theta_{s+t} - \bar{\theta})}{\sum_{s=1}^{S}(\theta_s - \bar{\theta})^2}$$

Remark 4.12.1. High autocorrelation implies slow mixing and inefficient sampling.

4.13 Diagnostics

- Trace plots
- Gelman-Rubin statistic
- Geweke test
- Effective sample size

Remark 4.13.1. No diagnostic guarantees convergence—only detects problems.

4.14 Summary

- Monte Carlo approximates integrals using random sampling
- Importance sampling improves flexibility

- Accept-reject enables indirect sampling
- MCMC handles complex posterior distributions
- Efficiency depends on mixing and autocorrelation

Chapter 5

Normal Model

5.1 The Normal Sampling Model

Definition 5.1.1. A random variable Y follows a normal distribution if:

$$Y \mid \theta, \sigma^2 \sim \mathcal{N}(\theta, \sigma^2)$$

with density:

$$p(y \mid \theta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y-\theta)^2\right)$$

Proposition 5.1.1. *For $Y \sim \mathcal{N}(\theta, \sigma^2)$:*

$$\mathbb{E}[Y] = \theta, \quad \mathrm{Var}(Y) = \sigma^2$$

Remark 5.1.1. The normal distribution is symmetric with mean = median = mode, and approximately 95% of mass lies within $\theta \pm 1.96\sigma$.

5.2 Likelihood Function

Let $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\theta, \sigma^2)$. Then:

$$p(y \mid \theta, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\theta)^2\right)$$

Remark 5.2.1. Inference can be separated into:

- inference on θ given σ^2
- inference on σ^2

5.3 Known Variance: Conjugate Analysis

5.3.1 Prior

Definition 5.3.1. Assume:

$$\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$$

5.3.2 Posterior

Theorem 5.3.1. *If* $Y_i \overset{iid}{\sim} \mathcal{N}(\theta, \sigma^2)$ *and* $\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$*, then:*

$$\theta \mid y \sim \mathcal{N}(\mu_n, \tau_n^2)$$

where:

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$$

$$\mu_n = \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{n}{\sigma^2}\bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}$$

Proof. Posterior:

$$p(\theta \mid y) \propto p(\theta) \cdot p(y \mid \theta)$$

Combine exponent terms:

$$-\frac{1}{2\tau_0^2}(\theta - \mu_0)^2 - \frac{1}{2\sigma^2}\sum(y_i - \theta)^2$$

Expand and collect quadratic terms in θ, yielding a normal kernel. □

5.3.3 Interpretation

Remark 5.3.1. The posterior mean is a weighted average:

$$\mu_n = \frac{\kappa_0}{\kappa_0 + n}\mu_0 + \frac{n}{\kappa_0 + n}\bar{y}$$

where $\kappa_0 = \sigma^2/\tau_0^2$.

Remark 5.3.2. Precision (inverse variance) is additive:

$$\text{precision}_{posterior} = \text{precision}_{prior} + \text{precision}_{data}$$

5.4 Unknown Variance: Joint Inference

Remark 5.4.1. When σ^2 is unknown, we introduce a prior on variance.

5.4.1 Prior Specification

Definition 5.4.1. Use:

$$\theta \mid \sigma^2 \sim \mathcal{N}(\mu_0, \sigma^2/\kappa_0)$$
$$\sigma^2 \sim \text{Inv-Gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0\sigma_0^2}{2}\right)$$

5.4.2 Posterior Updates

Proposition 5.4.1. *Posterior parameters:*

$$\kappa_n = \kappa_0 + n$$
$$\mu_n = \frac{\kappa_0\mu_0 + n\bar{y}}{\kappa_n}$$
$$\nu_n = \nu_0 + n$$
$$\sigma_n^2 = \frac{1}{\nu_n}\left[\nu_0\sigma_0^2 + (n-1)s^2 + \frac{\kappa_0 n}{\kappa_n}(\bar{y} - \mu_0)^2\right]$$

Remark 5.4.2. This is often summarized as: "posterior sum of squares = prior sum + data sum + correction term."

5.5 Conditional Posteriors

Proposition 5.5.1.

$$\theta \mid \sigma^2, y \sim \mathcal{N}(\mu_n, \sigma^2/\kappa_n)$$
$$\sigma^2 \mid y \sim \text{Inv-Gamma}\left(\frac{\nu_n}{2}, \frac{\nu_n\sigma_n^2}{2}\right)$$

5.6 Marginal Posterior for the Mean

Theorem 5.6.1. *The marginal posterior of θ is:*

$$\frac{\theta - \mu_n}{\sigma_n/\sqrt{\kappa_n}} \sim t_{\nu_n}$$

Remark 5.6.1. The t-distribution arises from integrating out uncertainty in σ^2.

5.7 Example: Midge Wing Length

Example 5.7.1. Data:

$$y = (1.64, 1.70, 1.72, 1.74, 1.82, 1.82, 1.82, 1.90, 2.08)$$

Summary:

$$\bar{y} = 1.804, \quad s^2 \approx 0.017$$

Using prior $\mu_0 = 1.9$, $\kappa_0 = 1$, we obtain:

$$\mu_n \approx 1.814, \quad \sigma_n^2 \approx 0.015$$

5.8 Sampling from the Posterior

Remark 5.8.1. We can sample from the joint posterior using:

1. Sample $\sigma^2 \sim$ Inv-Gamma
2. Sample $\theta \sim \mathcal{N}$

5.9 Bias and Mean Squared Error

Definition 5.9.1. Bias:

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

Definition 5.9.2. Mean Squared Error:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}^2$$

Remark 5.9.1. Bayesian estimators are typically biased but can have lower MSE due to shrinkage.

5.10 Exponential Family Representation

Proposition 5.10.1. *The normal distribution belongs to the exponential family:*

$$p(y \mid \theta, \sigma^2) = \exp\left(\frac{\theta}{\sigma^2}y - \frac{1}{2\sigma^2}y^2 - A(\theta, \sigma^2)\right)$$

Remark 5.10.1. Sufficient statistics:

$$t(y) = (y, y^2)$$

5.11 Summary

- Normal model is central to Bayesian data analysis
- Conjugate priors yield closed-form posteriors
- Posterior mean is a weighted average of prior and data
- Unknown variance leads to t-distributed marginal posterior
- Joint inference handled via conditional distributions

Chapter 6

Multinomial Model, Bayes Factors, and Indirect Methods

6.1 Multinomial Model

6.1.1 Model Setup

Definition 6.1.1. Let:

$$x = (x_1, \ldots, x_k), \quad \sum_{j=1}^{k} x_j = n$$

Then the multinomial model is:

$$p(x \mid \theta) = \prod_{j=1}^{k} \theta_j^{x_j}$$

where:

$$\theta = (\theta_1, \ldots, \theta_k), \quad \sum_{j=1}^{k} \theta_j = 1$$

Remark 6.1.1. This generalizes the binomial model to k categories.

6.1.2 Dirichlet Prior

Definition 6.1.2. The conjugate prior is:

$$\theta \sim \text{Dirichlet}(\alpha)$$

with density:

$$\pi(\theta) = \frac{1}{B(\alpha)} \prod_{j=1}^{k} \theta_j^{\alpha_j - 1}$$

Remark 6.1.2.

- α_j acts like prior counts

- $\alpha_j = 1$ gives a uniform prior
- when $k = 2$, Dirichlet reduces to Beta

6.1.3 Posterior

Theorem 6.1.1. *Given data x, the posterior is:*

$$\theta \mid x \sim \text{Dirichlet}(\alpha + x)$$

Proof.

$$p(\theta \mid x) \propto \prod_{j=1}^{k} \theta_j^{x_j} \cdot \prod_{j=1}^{k} \theta_j^{\alpha_j - 1} = \prod_{j=1}^{k} \theta_j^{x_j + \alpha_j - 1}$$

□

6.1.4 Interpretation

Remark 6.1.3. Posterior = prior counts + observed counts:

$$\alpha_j \to \alpha_j + x_j$$

Remark 6.1.4. Dirichlet provides a natural Bayesian smoothing for categorical probabilities.

6.2 Bayes Factors

6.2.1 Definition

Definition 6.2.1. Given two models M_1 and M_2:

$$\text{BF} = \frac{p(M_1 \mid x)}{p(M_2 \mid x)} \cdot \frac{p(M_2)}{p(M_1)} = \frac{p(x \mid M_1)}{p(x \mid M_2)}$$

Remark 6.2.1. Bayes factor compares how well each model explains the data.

6.2.2 Marginal Likelihood

Definition 6.2.2.

$$p(x \mid M) = \int p(x \mid \theta, M) \pi(\theta \mid M) \, d\theta$$

Remark 6.2.2. This integrates out parameter uncertainty.

6.2.3 Interpretation

Remark 6.2.3. Typical interpretation scales (Jeffreys):

- $1 \leq BF < 3$: weak evidence
- $3 \leq BF < 10$: substantial
- $10 \leq BF < 100$: strong
- $BF \geq 100$: decisive

6.2.4 Important Caveats

Remark 6.2.4.

- Bayes factors require **proper priors**
- If prior is improper $\Rightarrow$ marginal likelihood undefined
- Sensitive to prior specification

6.2.5 Frequentist vs Bayesian Testing

Remark 6.2.5.

- Frequentist: reject/not reject (no strength)
- Bayesian: provides strength of evidence via BF

6.3 Indirect Methods (Monte Carlo)

6.3.1 Motivation

Remark 6.3.1. We often need to compute:

$$\mathbb{E}[f(\theta)] = \int f(\theta)p(\theta)\,d\theta$$

but cannot evaluate the integral analytically.

6.4 Importance Sampling

6.4.1 Core Idea

Definition 6.4.1. Rewrite expectation using proposal $g(x)$:

$$\mathbb{E}[h(x)] = \int h(x)f(x)\,dx = \int h(x)\frac{f(x)}{g(x)}g(x)\,dx$$

6.4.2 Estimator

Definition 6.4.2. Unnormalized:

$$\hat{\mu}^* = \frac{1}{n}\sum_{i=1}^{n} h(X_i)\frac{f(X_i)}{g(X_i)}$$

Normalized:

$$\hat{\mu} = \sum_{i=1}^{n} h(X_i)w_i$$

where:

$$w_i = \frac{f(X_i)/g(X_i)}{\sum_j f(X_j)/g(X_j)}$$

6.4.3 Effective Sample Size

Definition 6.4.3.

$$\text{ESS} = \frac{n}{1 + \widehat{\text{Var}}(w^*)}$$

Remark 6.4.1. High variance weights $\Rightarrow$ low efficiency.

6.5 Accept-Reject Sampling

6.5.1 Algorithm

---

1. Choose proposal $g(x)$ and constant M such that:

$$f(x) \leq Mg(x)$$

2. Sample $X \sim g(x)$
3. Sample $U \sim \text{Uniform}(0, 1)$
4. Accept if:

$$U \leq \frac{f(X)}{Mg(X)}$$

6.5.2 Remarks

- Smaller $M \Rightarrow$ higher efficiency
- g must be easy to sample

6.6 Monte Carlo Approximation

Definition 6.6.1. Given samples $\theta^{(1)}, \ldots, \theta^{(S)}$:

$$\hat{\theta} = \frac{1}{S} \sum_{s=1}^{S} \theta^{(s)}$$

Definition 6.6.2. Monte Carlo variance:

$$\widehat{\text{Var}} = \frac{1}{S-1} \sum (\theta^{(s)} - \bar{\theta})^2$$

Remark 6.6.1. Accuracy improves with S by law of large numbers.

6.7 Summary

- Multinomial + Dirichlet gives clean conjugate updates
- Bayes factors compare models via marginal likelihoods
- Indirect methods approximate integrals when closed forms fail
- Importance sampling and AR are foundational tools before MCMC

Chapter 7

Markov Chain Monte Carlo, Gibbs Sampling, and Diagnostics

7.1 Motivation

Remark 7.1.1. In many Bayesian models, the posterior

$$p(\theta \mid y)$$

cannot be computed in closed form.

We instead approximate expectations:

$$\mathbb{E}[h(\theta) \mid y] = \int h(\theta) p(\theta \mid y)\, d\theta$$

using simulation methods.

7.2 Markov Chains

7.2.1 Definition

Definition 7.2.1. A sequence $\{X_t\}$ is a Markov chain if:

$$\mathbb{P}(X_{t+1} \mid X_t, \ldots, X_0) = \mathbb{P}(X_{t+1} \mid X_t)$$

7.2.2 Key Properties

- Irreducibility: all states communicate
- Aperiodicity: returns do not occur in fixed cycles
- Positive recurrence: expected return time is finite

Definition 7.2.2. A chain satisfying these properties is called ergodic.

7.2.3 Stationary Distribution

Definition 7.2.3. A distribution π is stationary if:

$$\pi = \pi P$$

Remark 7.2.1. MCMC constructs a Markov chain whose stationary distribution is the posterior:

$$\pi(\theta) = p(\theta \mid y)$$

7.3 Markov Chain Monte Carlo

Definition 7.3.1. MCMC generates dependent samples:

$$\theta^{(1)}, \theta^{(2)}, \dots$$

such that:

$$\theta^{(t)} \xrightarrow{d} p(\theta \mid y)$$

Proposition 7.3.1. *For an ergodic chain:*

$$\mathbb{E}[h(\theta) \mid y] \approx \frac{1}{T} \sum_{t=1}^{T} h(\theta^{(t)})$$

7.4 Metropolis-Hastings Algorithm

Definition 7.4.1. Given a proposal $q(\theta^* \mid \theta)$, define:

$$\alpha(\theta, \theta^*) = \min \left\{ \frac{p(\theta^* \mid y) q(\theta \mid \theta^*)}{p(\theta \mid y) q(\theta^* \mid \theta)}, 1 \right\}$$

---

1. Initialize $\theta^{(0)}$
2. For $t = 1, \dots, T$:
   (a) Sample $\theta^* \sim q(\cdot \mid \theta^{(t-1)})$
   (b) Accept with probability α
   (c) Otherwise set $\theta^{(t)} = \theta^{(t-1)}$

Remark 7.4.1. If q is symmetric, the ratio simplifies to:

$$\alpha = \min \left\{ \frac{p(\theta^* \mid y)}{p(\theta \mid y)}, 1 \right\}$$

7.5 Random Walk Metropolis

Definition 7.5.1. A common proposal is:

$$\theta^* = \theta^{(t-1)} + \epsilon$$

Remark 7.5.1. The choice of proposal variance determines:

- step size
- acceptance rate
- efficiency

7.6 Gibbs Sampling

7.6.1 Definition

Definition 7.6.1. Let $\theta = (\theta_1, \ldots, \theta_k)$. Gibbs sampling iteratively samples:

$$\begin{aligned}
\theta_1^{(t)} &\sim p(\theta_1 \mid \theta_2^{(t-1)}, \ldots, \theta_k^{(t-1)}, y) \\
\theta_2^{(t)} &\sim p(\theta_2 \mid \theta_1^{(t)}, \ldots, \theta_k^{(t-1)}, y) \\
&\vdots \\
\theta_k^{(t)} &\sim p(\theta_k \mid \theta_1^{(t)}, \ldots, \theta_{k-1}^{(t)}, y)
\end{aligned}$$

Remark 7.6.1. Gibbs sampling avoids rejection by sampling directly from full conditional distributions.

7.6.2 Example: Normal Model

- $\theta \mid \sigma^2, y \sim \mathcal{N}(\mu_n, \sigma^2/\kappa_n)$
- $\sigma^2 \mid \theta, y \sim \text{Inv-Gamma}(\cdot)$

Remark 7.6.2. Alternating these steps produces samples from the joint posterior.

7.7 Burn-in

Definition 7.7.1. Discard the first B samples:

$$\{\theta^{(B+1)}, \ldots, \theta^{(T)}\}$$

Remark 7.7.1. Initial samples depend on the starting value and may not represent the posterior.

7.8 Autocorrelation

Definition 7.8.1. Lag-k autocorrelation:

$$\rho_k = \frac{\sum_{t=1}^{T-k}(\theta^{(t)} - \bar{\theta})(\theta^{(t+k)} - \bar{\theta})}{\sum_{t=1}^{T}(\theta^{(t)} - \bar{\theta})^2}$$

Remark 7.8.1. High autocorrelation implies slow exploration of the parameter space.

7.9 Effective Sample Size

Definition 7.9.1.

$$\text{ESS} = \frac{T}{1 + 2\sum_{k=1}^{\infty} \rho_k}$$

Remark 7.9.1. ESS measures the number of independent samples equivalent to the correlated chain.

7.10 Trace Plots

Remark 7.10.1. Plotting $\theta^{(t)}$ against t provides a visual diagnostic:

- stable band indicates convergence
- trends or clustering indicate poor mixing

7.11 Formal Diagnostics

- Geweke diagnostic: compares early and late means
- Gelman-Rubin: compares multiple chains
- Heidelberger-Welch: tests stationarity
- Raftery-Lewis: estimates required sample size

Remark 7.11.1. Diagnostics can detect lack of convergence but cannot guarantee convergence.

7.12 Mixing

Definition 7.12.1. Mixing describes how quickly the chain explores the parameter space.

Remark 7.12.1.

- fast mixing: low autocorrelation
- slow mixing: high autocorrelation

7.13 Improving MCMC Performance

- run longer chains
- tune proposal distributions
- reparameterize the model
- block correlated variables
- introduce auxiliary variables

7.14 Summary

- MCMC approximates posterior expectations using dependent samples
- Metropolis-Hastings is general-purpose
- Gibbs sampling uses conditional structure
- diagnostics are essential for assessing convergence
- efficiency depends on mixing and autocorrelation

Chapter 8

Multivariate Normal Model

8.1 Multivariate Normal Distribution

Definition 8.1.1. A random vector $Y \in \mathbb{R}^p$ follows a multivariate normal distribution if:

$$Y \mid \theta, \Sigma \sim \mathcal{N}_p(\theta, \Sigma)$$

with density:

$$p(y \mid \theta, \Sigma) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(y-\theta)^\top \Sigma^{-1}(y-\theta)\right)$$

Proposition 8.1.1.

$$\mathbb{E}[Y] = \theta, \quad \mathrm{Var}(Y) = \Sigma$$

Remark 8.1.1. The covariance matrix Σ must be symmetric positive definite.

8.2 Sampling Model

Let $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}_p(\theta, \Sigma)$.

Proposition 8.2.1. *The likelihood is:*

$$p(y \mid \theta, \Sigma) \propto |\Sigma|^{-n/2} \exp\left(-\frac{1}{2}\sum_{i=1}^{n}(y_i-\theta)^\top \Sigma^{-1}(y_i-\theta)\right)$$

Remark 8.2.1. The sufficient statistics are:

$$\bar{y} = \frac{1}{n}\sum y_i, \quad S = \sum (y_i - \bar{y})(y_i - \bar{y})^\top$$

8.3 Known Covariance Matrix

8.3.1 Prior

Definition 8.3.1.

$$\theta \sim \mathcal{N}_p(\mu_0, \Lambda_0)$$

8.3.2 Posterior

Theorem 8.3.1.

$$\theta \mid y, \Sigma \sim \mathcal{N}_p(\mu_n, \Lambda_n)$$

where:

$$\Lambda_n = \left(\Lambda_0^{-1} + n\Sigma^{-1}\right)^{-1}$$

$$\mu_n = \Lambda_n \left(\Lambda_0^{-1}\mu_0 + n\Sigma^{-1}\bar{y}\right)$$

Remark 8.3.1. Posterior precision:

$$\Lambda_n^{-1} = \Lambda_0^{-1} + n\Sigma^{-1}$$

Remark 8.3.2. Posterior mean is a weighted average of prior mean and sample mean.

8.4 Unknown Covariance Matrix

8.4.1 Inverse-Wishart Prior

Definition 8.4.1.

$$\Sigma \sim \text{Inv-Wishart}(\nu_0, S_0)$$

Remark 8.4.1. This is the multivariate analogue of the inverse-gamma distribution.

8.4.2 Joint Prior

Definition 8.4.2.

$$\theta \mid \Sigma \sim \mathcal{N}_p(\mu_0, \Sigma/\kappa_0)$$

$$\Sigma \sim \text{Inv-Wishart}(\nu_0, S_0)$$

8.5 Posterior Distribution

Proposition 8.5.1. *Posterior parameters:*

$$\kappa_n = \kappa_0 + n$$

$$\mu_n = \frac{\kappa_0\mu_0 + n\bar{y}}{\kappa_n}$$

$$\nu_n = \nu_0 + n$$
$$S_n = S_0 + \sum_{i=1}^{n}(y_i - \bar{y})(y_i - \bar{y})^\top + \frac{\kappa_0 n}{\kappa_n}(\bar{y} - \mu_0)(\bar{y} - \mu_0)^\top$$

Proposition 8.5.2.

$$\theta \mid \Sigma, y \sim \mathcal{N}_p(\mu_n, \Sigma/\kappa_n)$$
$$\Sigma \mid y \sim \textit{Inv-Wishart}(\nu_n, S_n)$$

8.6 Gibbs Sampling for MVN Model

---

1. Initialize $\theta^{(0)}, \Sigma^{(0)}$
2. For $t = 1, \ldots, T$:
   (a) Sample:
   $$\theta^{(t)} \sim \mathcal{N}_p(\mu_n, \Sigma^{(t-1)}/\kappa_n)$$
   (b) Sample:
   $$\Sigma^{(t)} \sim \text{Inv-Wishart}(\nu_n, S_n(\theta^{(t)}))$$

Remark 8.6.1. This alternates between conditional distributions, producing samples from the joint posterior.

8.7 Wishart Distribution

Definition 8.7.1. The Wishart distribution is:

$$W \sim \text{Wishart}_p(\nu, \Sigma)$$

Proposition 8.7.1.

$$\mathbb{E}[W] = \nu\Sigma$$

Remark 8.7.1. The Wishart distribution is the distribution of a sample covariance matrix.

8.8 Inverse-Wishart Distribution

Proposition 8.8.1. *If* $\Sigma \sim \textit{Inv-Wishart}(\nu, S)$*:*

$$\mathbb{E}[\Sigma] = \frac{S}{\nu - p - 1}$$

Remark 8.8.1. Defined only when $\nu > p + 1$.

8.9 Multivariate t-Distribution

Proposition 8.9.1. *Marginalizing Σ yields:*

$$\theta \mid y \sim t_{\nu_n - p + 1}(\mu_n, S_n)$$

Remark 8.9.1. The multivariate t distribution accounts for uncertainty in covariance estimation.

8.10 Simulation of MVN

Proposition 8.10.1. *If $Z \sim \mathcal{N}_p(0, I)$ and $\Sigma = LL^\top$, then:*

$$Y = \theta + LZ \sim \mathcal{N}_p(\theta, \Sigma)$$

8.11 Summary

- Multivariate normal generalizes the normal model to vector-valued data
- Mean becomes a vector, variance becomes a covariance matrix
- Conjugate prior: Normal-Inverse-Wishart
- Posterior updates mirror univariate case using matrix algebra
- Gibbs sampling alternates between mean and covariance updates

Chapter 9

Group Comparisons and Hierarchical Models

9.1 Motivation

Remark 9.1.1. Many datasets are naturally grouped:

- Students within schools
- Patients within hospitals
- Firms within industries

Remark 9.1.2. We want to model:

- Within-group variability
- Between-group variability

9.2 Two-Group Comparison Model

9.2.1 Model Specification

Definition 9.2.1. For two groups:

$$Y_{i,1} = \mu + \delta + \epsilon_{i,1}$$
$$Y_{i,2} = \mu - \delta + \epsilon_{i,2}$$
$$\epsilon_{i,j} \sim \mathcal{N}(0, \sigma^2)$$

Remark 9.2.1.

- μ = overall mean
- δ = group difference
- σ^2 = shared variance

9.3 Priors

Definition 9.3.1.

$$\mu \sim \mathcal{N}(\mu_0, \gamma_0^2)$$
$$\delta \sim \mathcal{N}(\delta_0, \tau_0^2)$$
$$\sigma^2 \sim \text{Inv-Gamma}(\nu_0/2, \nu_0\sigma_0^2/2)$$

9.4 Posterior Conditionals

9.4.1 Posterior of μ

Proposition 9.4.1.

$$\mu \mid \delta, \sigma^2, y \sim \mathcal{N}(\mu_n, \gamma_n^2)$$

$$\gamma_n^2 = \left(\frac{1}{\gamma_0^2} + \frac{n_1 + n_2}{\sigma^2}\right)^{-1}$$

$$\mu_n = \gamma_n^2 \left[\frac{\mu_0}{\gamma_0^2} + \frac{\sum(y_{i,1} - \delta)}{\sigma^2} + \frac{\sum(y_{i,2} + \delta)}{\sigma^2}\right]$$

9.4.2 Posterior of δ

Proposition 9.4.2.

$$\delta \mid \mu, \sigma^2, y \sim \mathcal{N}(\delta_n, \tau_n^2)$$

$$\tau_n^2 = \left(\frac{1}{\tau_0^2} + \frac{n_1 + n_2}{\sigma^2}\right)^{-1}$$

$$\delta_n = \tau_n^2 \left[\frac{\delta_0}{\tau_0^2} + \frac{\sum(y_{i,1} - \mu)}{\sigma^2} - \frac{\sum(y_{i,2} - \mu)}{\sigma^2}\right]$$

9.4.3 Posterior of σ^2

Proposition 9.4.3.

$$\sigma^2 \mid \mu, \delta, y \sim \textit{Inv-Gamma}\left(\frac{\nu_n}{2}, \frac{\nu_n\sigma_n^2}{2}\right)$$

$$\nu_n = \nu_0 + n_1 + n_2$$

$$\nu_n\sigma_n^2 = \nu_0\sigma_0^2 + \sum(y_{i,1} - (\mu + \delta))^2 + \sum(y_{i,2} - (\mu - \delta))^2$$

9.5 Gibbs Sampling Algorithm

---

1. Initialize $\mu^{(0)}, \delta^{(0)}, \sigma^{2(0)}$
2. For $t = 1, \ldots, T$:
   (a) Sample:
   $$\mu^{(t)} \sim p(\mu \mid \delta^{(t-1)}, \sigma^{2(t-1)}, y)$$
   (b) Sample:
   $$\delta^{(t)} \sim p(\delta \mid \mu^{(t)}, \sigma^{2(t-1)}, y)$$
   (c) Sample:
   $$\sigma^{2(t)} \sim p(\sigma^2 \mid \mu^{(t)}, \delta^{(t)}, y)$$

9.6 Exchangeability

Definition 9.6.1. A sequence $Y_1, \ldots, Y_n$ is exchangeable if:

$$p(y_1, \ldots, y_n) = p(y_{\pi(1)}, \ldots, y_{\pi(n)})$$

for any permutation π.

Remark 9.6.1. Exchangeability justifies modeling observations as conditionally IID.

9.7 Hierarchical Normal Model

9.7.1 Model Structure

Definition 9.7.1. Within-group model:

$$Y_{ij} \mid \theta_j, \sigma^2 \sim \mathcal{N}(\theta_j, \sigma^2)$$

Between-group model:

$$\theta_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, \tau^2)$$

Remark 9.7.1.

- θ_j = group-specific mean
- μ = global mean
- τ^2 = between-group variance

9.8 Interpretation

Remark 9.8.1. This model induces **partial pooling**:

- Estimates are shrunk toward the global mean
- Amount of shrinkage depends on τ^2

9.9 Shrinkage Effect

Proposition 9.9.1. *Posterior mean of θ_j is:*

$$\mathbb{E}[\theta_j \mid y] = w_j \bar{y}_j + (1 - w_j)\mu$$

Remark 9.9.1.

- If τ^2 is small $\rightarrow$ strong pooling
- If τ^2 is large $\rightarrow$ weak pooling

9.10 Why Hierarchical Models Matter

- Avoid overfitting small groups
- Borrow strength across groups
- Naturally quantify uncertainty
- Handle missing data via latent variables

9.11 Summary

- Group comparison extends normal models to structured data
- Parameters become layered (data $\rightarrow$ group $\rightarrow$ population)
- Conjugacy enables efficient Gibbs sampling
- Hierarchical models introduce shrinkage and improve estimation stability

Chapter 10

Bayesian Linear Models

10.1 Model Setup

Definition 10.1.1. The Bayesian linear model is:

$$y = X\beta + \epsilon$$

$$\epsilon \sim \mathcal{N}(0, \sigma^2 I_n)$$

Remark 10.1.1.

- $y \in \mathbb{R}^n$
- $X \in \mathbb{R}^{n \times p}$
- $\beta \in \mathbb{R}^p$
- $\sigma^2 =$ noise variance

10.2 Likelihood

Proposition 10.2.1.

$$p(y \mid \beta, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta)\right)$$

10.3 Conjugate Prior

Definition 10.3.1. Normal–Inverse-Gamma prior:

$$\beta \mid \sigma^2 \sim \mathcal{N}(\beta_0, \sigma^2 V_0)$$

$$\sigma^2 \sim \text{Inv-Gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\right)$$

Remark 10.3.1.
- Prior scales with σ^2
- Enables closed-form posterior

10.4 Posterior Distribution

10.4.1 Posterior of β

Proposition 10.4.1.

$$\beta \mid \sigma^2, y \sim \mathcal{N}(\beta_n, \sigma^2 V_n)$$

$$V_n = (V_0^{-1} + X^\top X)^{-1}$$

$$\beta_n = V_n(V_0^{-1}\beta_0 + X^\top y)$$

10.4.2 Posterior of σ^2

Proposition 10.4.2.

$$\sigma^2 \mid y \sim \textit{Inv-Gamma}\left(\frac{\nu_n}{2}, \frac{\nu_n \sigma_n^2}{2}\right)$$

$$\nu_n = \nu_0 + n$$

$$\nu_n \sigma_n^2 = \nu_0 \sigma_0^2 + (y - X\beta_n)^\top (y - X\beta_n) + (\beta_n - \beta_0)^\top V_0^{-1} (\beta_n - \beta_0)$$

10.5 Marginal Posterior of β

Proposition 10.5.1.

$$\beta \mid y \sim t_{\nu_n}(\beta_n, \sigma_n^2 V_n)$$

Remark 10.5.1.
- arises from integrating out σ^2
- heavier tails than normal

10.6 Posterior Predictive Distribution

Proposition 10.6.1. *For a new observation y^* with covariate x^*:*

$$y^* \mid y \sim t_{\nu_n}\left(x^{*\top}\beta_n, \sigma_n^2(1 + x^{*\top} V_n x^*)\right)$$

10.7 Interpretation

Remark 10.7.1. Posterior mean:

$$\beta_n = (V_0^{-1} + X^\top X)^{-1}(V_0^{-1}\beta_0 + X^\top y)$$

Remark 10.7.2. This is a weighted average of:

- Prior information (β_0)
- Data information ($X^\top y$)

10.8 Connection to OLS

Proposition 10.8.1. *If prior is diffuse:*

$$V_0^{-1} \to 0$$

then:

$$\beta_n \to (X^\top X)^{-1} X^\top y$$

Remark 10.8.1. Bayesian estimator reduces to OLS.

10.9 Shrinkage Interpretation

Remark 10.9.1. Bayesian regression performs shrinkage:

- pulls estimates toward β_0
- strength depends on V_0

10.10 Special Case: Ridge Regression

Proposition 10.10.1. *If:*

$$\beta \sim \mathcal{N}(0, \lambda^{-1} I)$$

then:

$$\beta_n = (X^\top X + \lambda I)^{-1} X^\top y$$

Remark 10.10.1. Bayesian interpretation of ridge regression.

1. Initialize $\beta^{(0)}, \sigma^{2(0)}$
2. For $t = 1, \ldots, T$:
   (a) Sample:
   $$\beta^{(t)} \sim p(\beta \mid \sigma^{2(t-1)}, y)$$
   (b) Sample:
   $$\sigma^{2(t)} \sim p(\sigma^2 \mid \beta^{(t)}, y)$$

10.11 Gibbs Sampling

10.12 Summary

- Linear model extends normal model to multivariate predictors
- Conjugate priors give closed-form posterior
- Posterior is Normal–Inverse-Gamma
- Marginal posterior is Student-t
- Includes OLS and ridge as special cases

Chapter 11

Generalized Linear Models (GLMs)

11.1 Motivation

Remark 11.1.1. Linear models assume:

- Gaussian noise
- Constant variance
- Unbounded response

These assumptions fail for:

- Binary data
- Count data
- Skewed / non-Gaussian responses

11.2 GLM Structure

Definition 11.2.1. A generalized linear model consists of:

1. Random component:
$$y_i \sim \text{Exponential Family}$$
2. Systematic component:
$$\eta_i = x_i^\top \beta$$
3. Link function:
$$g(\mu_i) = \eta_i$$

Remark 11.2.1.

- $\mu_i = \mathbb{E}[y_i \mid x_i]$

- $g(\cdot)$ connects mean to linear predictor

11.3 Exponential Family Form

Definition 11.3.1.

$$p(y \mid \theta) = h(y) \exp\{\phi(\theta)t(y) - A(\theta)\}$$

Remark 11.3.1. GLMs arise because many common distributions (Bernoulli, Poisson, Normal) belong to this family

11.4 Canonical Link

Definition 11.4.1. A canonical link satisfies:

$$\eta = \phi(\theta)$$

Remark 11.4.1. This simplifies likelihood and inference.

11.5 Common GLMs

11.5.1 Logistic Regression (Binary Data)

Definition 11.5.1.

$$y_i \sim \text{Bernoulli}(p_i)$$

$$\log\left(\frac{p_i}{1-p_i}\right) = x_i^\top \beta$$

Remark 11.5.1.

- Models probabilities in $(0, 1)$
- Uses logit link

Proposition 11.5.1. *Likelihood:*

$$p(y \mid \beta) = \prod_i p_i^{y_i}(1-p_i)^{1-y_i}$$

Remark 11.5.2. Appears in your notes as:

$$\text{logit}(p_i) = \alpha + \beta x_i$$

11.5.2 Poisson Regression (Count Data)

Definition 11.5.2.

$$y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = x_i^\top \beta$$

Remark 11.5.3.

- Ensures $\lambda_i > 0$
- Natural for count data

11.5.3 Gaussian GLM

Definition 11.5.3.

$$y_i \sim \mathcal{N}(x_i^\top \beta, \sigma^2)$$

Remark 11.5.4. This reduces to the standard linear model.

11.6 Bayesian GLMs

11.6.1 Prior

Definition 11.6.1.

$$\beta \sim \mathcal{N}(\beta_0, \Sigma_0)$$

11.7 Posterior

Proposition 11.7.1.

$$p(\beta \mid y) \propto p(y \mid \beta) p(\beta)$$

Remark 11.7.1.

- No closed form (non-conjugate)
- Requires numerical methods

11.8 Inference Methods

11.8.1 Laplace Approximation

Definition 11.8.1. Approximate posterior by Gaussian around mode:

$$p(\beta \mid y) \approx \mathcal{N}(\hat{\beta}, H^{-1})$$

Remark 11.8.1.

- $\hat{\beta}$ = MAP estimate
- H = Hessian of log posterior

11.8.2 Metropolis-Hastings

1. Propose $\beta^* \sim q(\cdot \mid \beta^{(t)})$
2. Accept with probability:

$$\min\left(1, \frac{p(y \mid \beta^*)p(\beta^*)}{p(y \mid \beta^{(t)})p(\beta^{(t)})}\right)$$

Remark 11.8.2. Used when posterior cannot be sampled directly

11.8.3 Gibbs Sampling (Augmentation)

Remark 11.8.3.

- Introduce latent variables
- Convert model into conditionally conjugate form

11.9 Interpretation

Remark 11.9.1.

- Coefficients act on transformed mean
- Interpretation depends on link function

11.9.1 Logistic Regression

Proposition 11.9.1.

$$\beta_j = \textit{log-odds change per unit increase in } x_j$$

11.9.2 Poisson Regression

Proposition 11.9.2.

$$e^{\beta_j} = \textit{multiplicative effect on expected count}$$

11.10 Summary

- GLMs extend linear models to non-Gaussian data
- Built from exponential family distributions
- Use link functions to ensure valid outputs
- Bayesian GLMs are typically non-conjugate
- Require approximation (Laplace, MCMC, VI)

Chapter 12

Bayesian Probit Regression Analysis

12.1 Model Setup

Definition 12.1.1. Binary response model:

$$y_i \in \{0, 1\}$$

$$\mathbb{P}(y_i = 1 \mid x_i) = \Phi(x_i^\top \beta)$$

Remark 12.1.1.

- $\Phi(\cdot)$ = standard normal CDF
- ensures probabilities lie in $(0, 1)$

12.2 Latent Variable Representation

Definition 12.2.1. Introduce latent variable:

$$z_i = x_i^\top \beta + \epsilon_i$$

$$\epsilon_i \sim \mathcal{N}(0, 1)$$

$$y_i = \begin{cases} 1 & \text{if } z_i > 0 \\ 0 & \text{if } z_i \leq 0 \end{cases}$$

Proposition 12.2.1.

$$\mathbb{P}(y_i = 1 \mid x_i) = \mathbb{P}(z_i > 0) = \Phi(x_i^\top \beta)$$

Remark 12.2.1. Probit regression is a thresholded Gaussian linear model.

12.3 Likelihood

Proposition 12.3.1.

$$p(y \mid \beta) = \prod_{i=1}^{n} \Phi(x_i^\top \beta)^{y_i} \left[1 - \Phi(x_i^\top \beta)\right]^{1-y_i}$$

Remark 12.3.1. No conjugate prior exists in this form.

12.4 Prior

Definition 12.4.1.

$$\beta \sim \mathcal{N}(\beta_0, \Sigma_0)$$

12.5 Posterior

Proposition 12.5.1.

$$p(\beta \mid y) \propto p(y \mid \beta) p(\beta)$$

Remark 12.5.1. Posterior has no closed form $\rightarrow$ requires sampling.

12.6 Data Augmentation

Definition 12.6.1. Introduce latent variables:

$$z_i \mid \beta \sim \mathcal{N}(x_i^\top \beta, 1)$$

Proposition 12.6.1. *Conditional distributions:*

$$z_i \mid y_i, \beta \sim \begin{cases} \mathcal{N}(x_i^\top \beta, 1) \text{ truncated at } (0, \infty) & y_i = 1 \\ \mathcal{N}(x_i^\top \beta, 1) \text{ truncated at } (-\infty, 0] & y_i = 0 \end{cases}$$

Remark 12.6.1. This converts the model into a Gaussian regression problem.

12.7 Conditional Posterior of β

Proposition 12.7.1.

$$\beta \mid z \sim \mathcal{N}(\beta_n, V_n)$$

$$V_n = (X^\top X + \Sigma_0^{-1})^{-1}$$

$$\beta_n = V_n (X^\top z + \Sigma_0^{-1} \beta_0)$$

Remark 12.7.1. Conditional on z, this is identical to Bayesian linear regression.

12.8 Gibbs Sampling Algorithm

---

1. Initialize $\beta^{(0)}$
2. For $t = 1, \ldots, T$:
   (a) Sample latent variables:
   $$z_i^{(t)} \sim p(z_i \mid y_i, \beta^{(t-1)})$$
   (b) Sample coefficients:
   $$\beta^{(t)} \sim p(\beta \mid z^{(t)})$$

Remark 12.8.1.
- Fully Gibbs $\rightarrow$ no MH needed
- Efficient and stable

12.9 Posterior Prediction

Proposition 12.9.1.
$$\mathbb{P}(y^* = 1 \mid x^*, \beta) = \Phi(x^{*\top}\beta)$$

Remark 12.9.1. Posterior predictive:

$$\mathbb{P}(y^* = 1 \mid y) = \int \Phi(x^{*\top}\beta) p(\beta \mid y)\, d\beta$$

12.10 Interpretation

Remark 12.10.1.
- β_j affects latent variable z
- not directly interpretable as odds (unlike logistic)

12.11 Comparison with Logistic Regression

Proposition 12.11.1.
- *Logistic: logit link*
- *Probit: normal CDF link*

Remark 12.11.1.
- Probit allows exact Gibbs via augmentation
- Logistic typically requires Metropolis or specialized augmentation

12.12 Summary

- Probit model uses Gaussian latent variable
- Binary outcomes arise via thresholding
- Data augmentation restores conjugacy
- Enables efficient Gibbs sampling
- Closely related to linear Gaussian models

Chapter 13

Missing Data and Imputation

13.1 Motivation

Remark 13.1.1. Real datasets are rarely complete.

Naive approaches:

- Delete missing observations (information loss)
- Fill with mean or constants (bias)

Bayesian approach:

- Treat missing values as *unknown random variables*
- Integrate or sample them as part of inference

13.2 Data Decomposition

Definition 13.2.1. Partition data:

$$Y = (Y_{\text{obs}}, Y_{\text{miss}})$$

Proposition 13.2.1. *Target posterior:*

$$p(\theta \mid Y_{obs}) = \int p(\theta, Y_{miss} \mid Y_{obs})\, dY_{miss}$$

Remark 13.2.1. Integration is usually intractable $\rightarrow$ use sampling.

13.3 Missing Data Mechanisms

Definition 13.3.1. MCAR (Missing Completely at Random):

$$p(R \mid Y_{\text{obs}}, Y_{\text{miss}}, \theta) = p(R)$$

MAR (Missing at Random):

$$p(R \mid Y_{\text{obs}}, Y_{\text{miss}}, \theta) = p(R \mid Y_{\text{obs}})$$

MNAR (Not Missing at Random): depends on Y_{miss}

Remark 13.3.1.

- MCAR and MAR $\rightarrow$ standard Bayesian imputation valid
- MNAR $\rightarrow$ must explicitly model missingness

13.4 Bayesian Formulation

Proposition 13.4.1. *Joint posterior:*

$$p(\theta, Y_{miss} \mid Y_{obs}) \propto p(Y_{obs}, Y_{miss} \mid \theta)\, p(\theta)$$

Remark 13.4.1. Missing values are treated exactly like latent variables.

13.5 Data Augmentation

Remark 13.5.1. Same idea as probit:

Unknown quantities $\rightarrow$ introduce $\rightarrow$ sample $\rightarrow$ iterate

Definition 13.5.1. Augmented sampling scheme:

- Treat Y_{miss} as parameters
- Sample them along with θ

13.6 Gibbs Sampling with Missing Data

Remark 13.6.1. This is a direct extension of:

- Probit latent variable sampling
- General Gibbs sampling

At iteration t:

1. Sample parameters:
$$\theta^{(t)} \sim p(\theta \mid Y_{\text{obs}}, Y_{\text{miss}}^{(t-1)})$$
2. Sample missing data:
$$Y_{\text{miss}}^{(t)} \sim p(Y_{\text{miss}} \mid Y_{\text{obs}}, \theta^{(t)})$$

13.7 Key Insight

Proposition 13.7.1. *Imputation is sampling from:*

$$p(Y_{miss} \mid Y_{obs}, \theta)$$

Remark 13.7.1.

- Not a single "best guess"
- A full distribution of plausible values

13.8 Example: Normal Model with Missing Data

Definition 13.8.1. Assume:

$$y_i \sim \mathcal{N}(\mu, \sigma^2)$$

Proposition 13.8.1. *Conditional imputation:*

$$y_i^{miss} \mid \mu, \sigma^2 \sim \mathcal{N}(\mu, \sigma^2)$$

Remark 13.8.1. Missing values are sampled exactly like observed ones.

13.9 Multiple Imputation

Definition 13.9.1. Instead of one dataset:

- Generate multiple completed datasets
- Analyze each
- combine results

Remark 13.9.1. Captures uncertainty from missingness.

13.10 Complete Case vs Bayesian Approach

Proposition 13.10.1.

- *Complete case:*

  - *biased unless MCAR*
  - *wastes data*
- *Bayesian:*
  - *uses all available information*
  - *propagates uncertainty*

13.11 Connection to Earlier Topics

Remark 13.11.1.

- Probit: latent z_i
- Missing data: latent Y_{miss}
- Both use:
  - augmentation
  - Gibbs sampling

13.12 General Template

1. Initialize missing values
2. For $t = 1, \ldots, T$:
   (a) Sample $\theta^{(t)}$
   (b) Sample $Y_{\text{miss}}^{(t)}$
3. Use samples for inference

13.13 Summary

- Missing data treated as latent variables
- Posterior integrates over missing uncertainty
- Gibbs sampling enables tractable inference
- Avoids bias from ad-hoc imputation
- Unifies with latent variable modeling

Chapter 14

GLM and Beyond

14.1 Limitations of GLMs

Remark 14.1.1. Generalized Linear Models assume:

- Correct link function
- Correct distributional form
- Independent observations

These assumptions may fail in practice.

14.2 Non-Conjugacy

Remark 14.2.1. For most GLMs:

- Posterior distributions are not available in closed form
- Conjugate priors do not exist

Proposition 14.2.1.

$$p(\beta \mid y) \propto p(y \mid \beta)p(\beta)$$

Remark 14.2.2. Requires approximation or simulation methods.

14.3 Laplace Approximation

Definition 14.3.1. Approximate posterior around mode:

$$p(\beta \mid y) \approx \mathcal{N}(\hat{\beta}, H^{-1})$$

Remark 14.3.1.

- $\hat{\beta}$ = MAP estimate

- H = Hessian of log posterior

14.4 Data Augmentation

Remark 14.4.1. Introduce latent variables to simplify models:

- Probit: latent Gaussian variable
- Missing data: latent observations

Remark 14.4.2. Transforms non-conjugate models into conditionally conjugate ones.

14.5 Hierarchical GLMs

Definition 14.5.1. Extend GLMs with group structure:

$$y_{ij} \mid \theta_j \sim \text{GLM}$$
$$\theta_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, \tau^2)$$

Remark 14.5.1. Allows:

- Partial pooling
- Modeling heterogeneity

14.6 Overdispersion

Remark 14.6.1. Poisson assumption:

$$\text{Var}(Y) = \mathbb{E}[Y]$$

Often violated in practice.

Proposition 14.6.1. *Alternative:*

- *Negative Binomial model*
- *Random effects models*

14.7 Robust Models

Remark 14.7.1. Normal errors are sensitive to outliers.

Alternatives:

- Student-t likelihood
- Heavy-tailed priors

14.8 Nonparametric Extensions

Remark 14.8.1. Relax parametric assumptions:

- Dirichlet Process models
- Mixture models

14.9 Summary

- GLMs extend linear models to non-Gaussian data
- Non-conjugacy requires approximation methods
- Data augmentation restores tractability
- Hierarchical extensions capture complex structure
- Robust and nonparametric models improve flexibility

Chapter 15

Model Selection

15.1 Motivation

Remark 15.1.1. Given competing models $M_1, M_2, \ldots$, we want to determine which model best explains the data.

15.2 Bayesian Model Comparison

Definition 15.2.1. Posterior model probability:

$$p(M_i \mid y) = \frac{p(y \mid M_i)p(M_i)}{\sum_j p(y \mid M_j)p(M_j)}$$

15.3 Marginal Likelihood

Definition 15.3.1.

$$p(y \mid M) = \int p(y \mid \theta, M)p(\theta \mid M)\, d\theta$$

Remark 15.3.1. Accounts for:

- Model fit
- Model complexity

15.4 Bayes Factors

Definition 15.4.1. For models M_1 and M_2:

$$BF_{12} = \frac{p(y \mid M_1)}{p(y \mid M_2)}$$

Remark 15.4.1.
- $BF > 1$ favors M_1
- $BF < 1$ favors M_2

15.5 Interpretation

Remark 15.5.1.
- $1 \leq BF < 3$: weak evidence
- $3 \leq BF < 10$: moderate evidence
- $10 \leq BF < 100$: strong evidence
- $BF \geq 100$: decisive

15.6 Important Considerations

Remark 15.6.1.
- Requires proper priors
- Sensitive to prior choice
- Penalizes model complexity automatically

15.7 Marginalization Paradox

Remark 15.7.1. If prior is improper:

$$p(y \mid M) \text{ is undefined}$$

Therefore Bayes factors cannot be computed.

15.8 Alternative Criteria

15.8.1 BIC

Definition 15.8.1.

$$\text{BIC} = -2 \log p(y \mid \hat{\theta}) + k \log n$$

Remark 15.8.1.
- Penalizes complexity
- Large sample approximation to Bayes factor

15.8.2 Deviance Information Criterion (DIC)

Definition 15.8.2.

$$\text{DIC} = \bar{D} + p_D$$

Remark 15.8.2.

- $\bar{D}$ = posterior mean deviance
- p_D = effective number of parameters

15.8.3 Log Pseudo-Marginal Likelihood (LPML)

Remark 15.8.3. Based on leave-one-out predictive densities.

15.9 Posterior Predictive Checks

Definition 15.9.1. Compare observed data to simulated data:

$$y^{rep} \sim p(y^{rep} \mid y)$$

Remark 15.9.1. Detects model misfit.

15.10 Tradeoff: Fit vs Complexity

Remark 15.10.1. Good models:

- Fit data well
- Avoid overfitting

15.11 Summary

- Model selection compares competing explanations of data
- Bayes factors use marginal likelihoods
- Proper priors are required
- Alternative criteria approximate model evidence
- Balance between fit and complexity is essential